

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #10
Processes and Layout
Design of Layouts
Charts

Ceyda Oğuz
coguz@ku.edu.tr
Koç University

1

Agenda of the Day

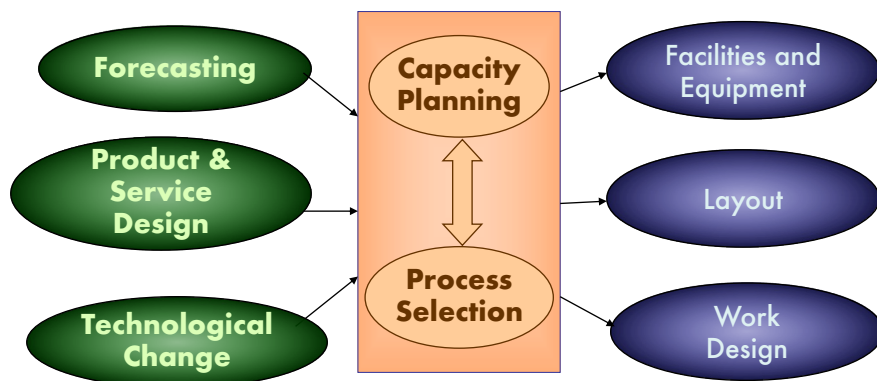
- Recap on processes and layout
- Design of layouts
- Tools we use to design layouts

INDR 100

2

2

Process Selection and System Design



Capacity is significantly impacted by process selection and facility layout.

INDR 100

3

3

Process Selection → Layout

- ❑ Production volume and product variety determine the type of the physical arrangement of the production system, called the **layout** of the system.
- ❑ **Production volume** refers to the number of units of a given part or product produced annually by the plant
- ❑ **Product variety** refers to the different product designs or types that are produced in a plant.

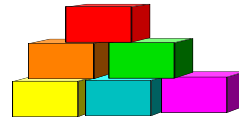
INDR 100

4

4

Facilities Layout

- ❑ The arrangement of departments, work centers, and equipment, with particular emphasis on movement of work (customers or materials) through the system.
- ❑ Objectives include:
 - quickest flow of material
 - lowest cost
 - least amount of handling in processing the product from the receipt of raw material to the delivery of the final product.



INDR 100

5

5

The Need for Layout Design Decisions

Inefficient operations

For Example:
High Cost
Bottlenecks



Changes in the design
of products or services

The introduction of new
products or services

Accidents



Safety hazards

7

7

The Need for Layout Design Decisions (Cont.)

Changes in environmental or other legal requirements



Changes in volume of output or mix of products

Changes in methods or equipment

Morale problems



8

8

Design of layouts

- ❑ Assume that we select the production process. How to proceed to design the layout?
- ❑ What do you need to know?
 - Understand how the production process evolves for all products.
 - Decide on the relative positions of the departments or lines or ... accordingly.

Charts and Diagrams...

INDR 100

9

9

Objectives of Charts and Diagrams

- ❑ Communication and documentation
- ❑ Use of algorithms
- ❑ Detailed analysis (divide & conquer)
- ❑ A structure for improvements
- ❑ Represent a proposed new work, process or method (when there is no current process or method)
 - Detail level for elements: to be decided
 - Additional information: to be decided

INDR 100

10

10

Charts

- ❑ Process (operations and flow) charts:
 - Material flow
 - Man flow
 - Information flow
 - Variations: with layouts
- ❑ From-to charts
- ❑ Activity relationship diagrams

INDR 371

INDR 100

12

12

Operations Process Chart

- ❑ Records the main activities of the process through the symbols of operation, storage and inspection only.
- ❑ This is used to present a birds eye view of the entire process
- ❑ They are used in drawing layout of plant or directing wasteful material or saving in machine tools

INDR 100

13

13

Flow Process Chart

- This chart uses all 5 activities for method analysis
- A single chart only records the activities of only one subject
- It is prepared for one component of an assembly at a time
- It is used for developing new method
- This chart gives information regarding distance moved, and time taken for different elements

INDR 100

14

14

Describing processes visually

□ Process charts: Operations process chart & Flow process chart (Man, Machine, Material)

– Symbols

- Operation 
- Storage 
- Inspection 
- Delay (flow process chart only) 
- Transportation (flow process chart only) 

INDR 100

15

15

Symbols and definitions

Sr. #	Symbol	Description
1		Operation: the item is intentionally changed in one of its characteristics, eg. Filling a bottle with soft drink, bending a metal sheet
2		Transportation: The item is moved from one place to another (except when the movement occurs as an integral part of an operation or inspection). Eg. Moving an item on conveyer belt b/w operations
3		Inspection: the unit at an inspection point is compared with the quality standard established for that point

INDR 100

16

16

Symbols and definitions

Sr. #	Symbol	Description
4		Delay: the next planned action does not take place. Eg: delay in moving a unit & delay in performance of the next operation
5		Storage: An item is stored in a place such its withdrawal requires authorization

INDR 100

17

17

How to prepare a process chart for a specific item

- ❑ Identify the parts required to produce the item (either purchased or manufactured)
- ❑ Determine the operations required to fabricate each part and sequence them
- ❑ Determine the sequence or assembly for purchased and manufactured parts
- ❑ Draw the process chart as explained
- ❑ Put time standards, operation numbers and descriptions
- ❑ Other relevant information can be added

INDR 100

18

18

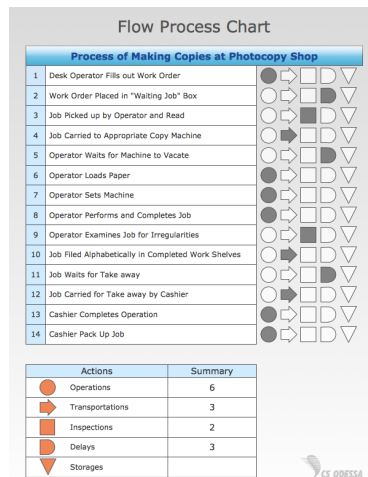


19



20

Examples

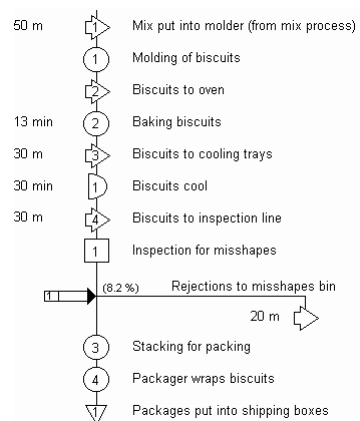


INDR 100

21

21

Examples



INDR 100

22

22

Questions when producing Process charts

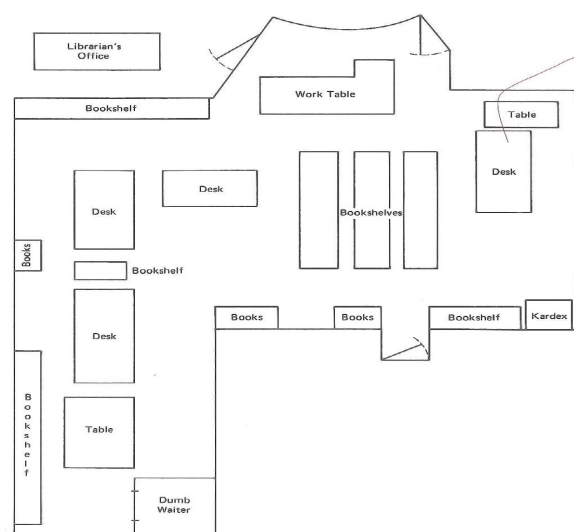
- ❑ Can I eliminate this step?
- ❑ Can I automate this step?
- ❑ Can I combine this step with another one?
- ❑ Can I change the routing to reduce distances travelled?
- ❑ Can I move workstations closer to each other?
- ❑ Can I justify production aids to increase effectiveness?
- ❑ How much does this part cost to produce?

INDR 100

23

23

Example: Library Layout



INDR 100

24

24

Example: processing new books

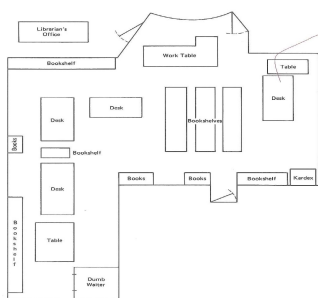
PRESENT METHOD ☒ PROPOSED METHOD ☐		FLOW PROCESS CHART		PAGE 1 OF 1	
PART NAME: <u>Book</u>		PROCESS DESCRIPTION: <u>Book Cataloging</u>		SUMMARY	
DEPARTMENT: <u>Library</u>		PLANT: <u>MV College</u>		NO.	
ANALYST: <u>LSA</u>		DATE: <u>4.1.201</u>		NOTE: SKETCHES ARE LOCATED ON BACK	
				TOTAL STEPS <u>13</u>	
				DISTANCE TRAVELED <u>165'</u>	
STEP	OPERATION TRANSPORTATION INSPECTION DELAY STORAGE	DESCRIPTION	TIME		
1	●○□□▽	Receive Books			
2	○●□□▽	Books to Check In	75		
3	●○□□▽	Books Checked In			
4	○●□□▽	Books to Research	30		
5	●○□□▽	Catalog Books			
6	○●□□▽	To Storage with No Numbers	20		
7	○●□□▽	Store			
8	○●□□▽	To Catalog When Number Available	20		
9	○●□□▽	To Verification	10		
10	○●□□▽	Verify			
11	○●□□▽	To Call Number Application	5		
12	○●□□▽	Apply Call Number			
13	○●□□▽	To Storage	5		
	○●□□▽				
	○●□□▽				
	○●□□▽				

INDR 100

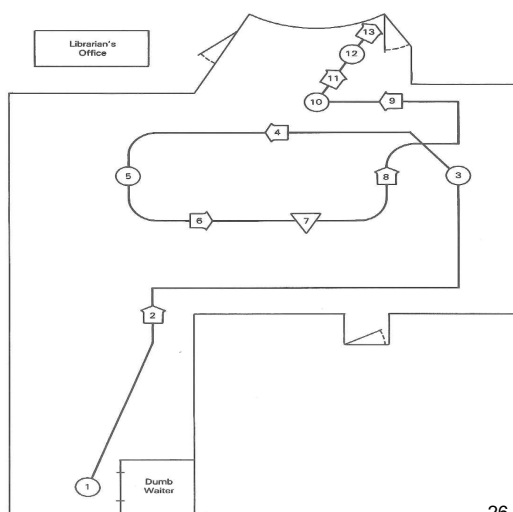
25

25

Example: layout flow chart



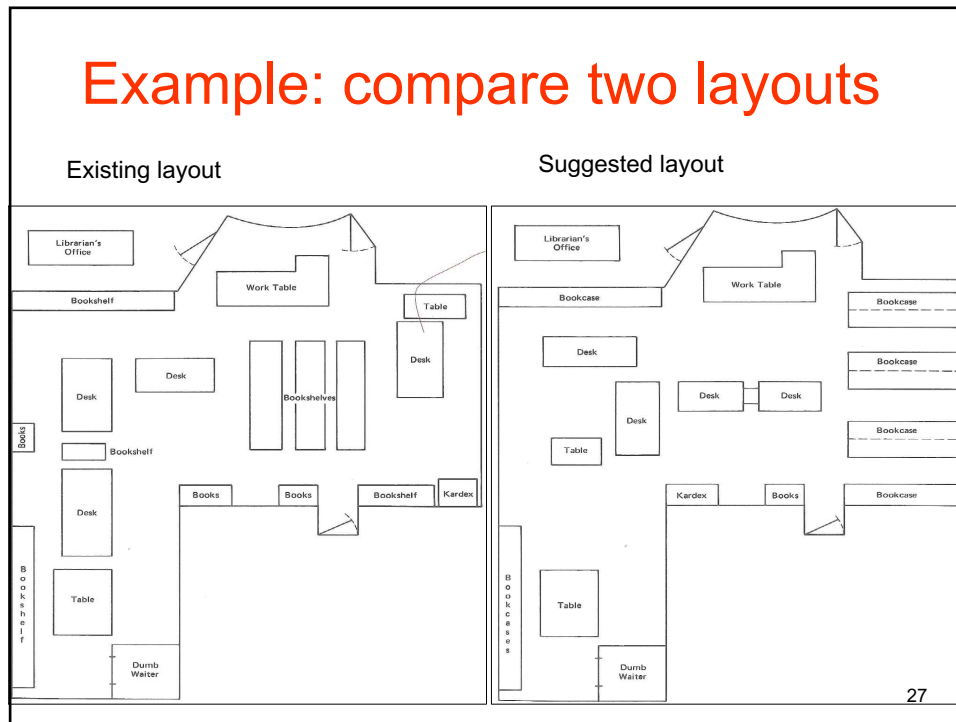
PRESENT METHOD ☒ PROPOSED METHOD ☐		FLOW PROCESS CHART		PAGE 1 OF 1	
PART NAME: <u>Book</u>		PROCESS DESCRIPTION: <u>Book Cataloging</u>		SUMMARY	
DEPARTMENT: <u>Library</u>		PLANT: <u>MV College</u>		NO.	
ANALYST: <u>LSA</u>		DATE: <u>4.1.201</u>		NOTE: SKETCHES ARE LOCATED ON BACK	
				TOTAL STEPS <u>13</u>	
				DISTANCE TRAVELED <u>165'</u>	
STEP	OPERATION TRANSPORTATION INSPECTION DELAY STORAGE	DESCRIPTION	TIME		
1	●○□□▽	Receive Books			
2	○●□□▽	Books to Check In	75		
3	●○□□▽	Books Checked In			
4	○●□□▽	Books to Research	30		
5	●○□□▽	Catalog Books			
6	○●□□▽	To Storage with No Numbers	20		
7	○●□□▽	Store			
8	○●□□▽	To Catalog When Number Available	20		
9	○●□□▽	To Verification	10		
10	○●□□▽	Verify			
11	○●□□▽	To Call Number Application	5		
12	○●□□▽	Apply Call Number			
13	○●□□▽	To Storage	5		
	○●□□▽				
	○●□□▽				
	○●□□▽				



26

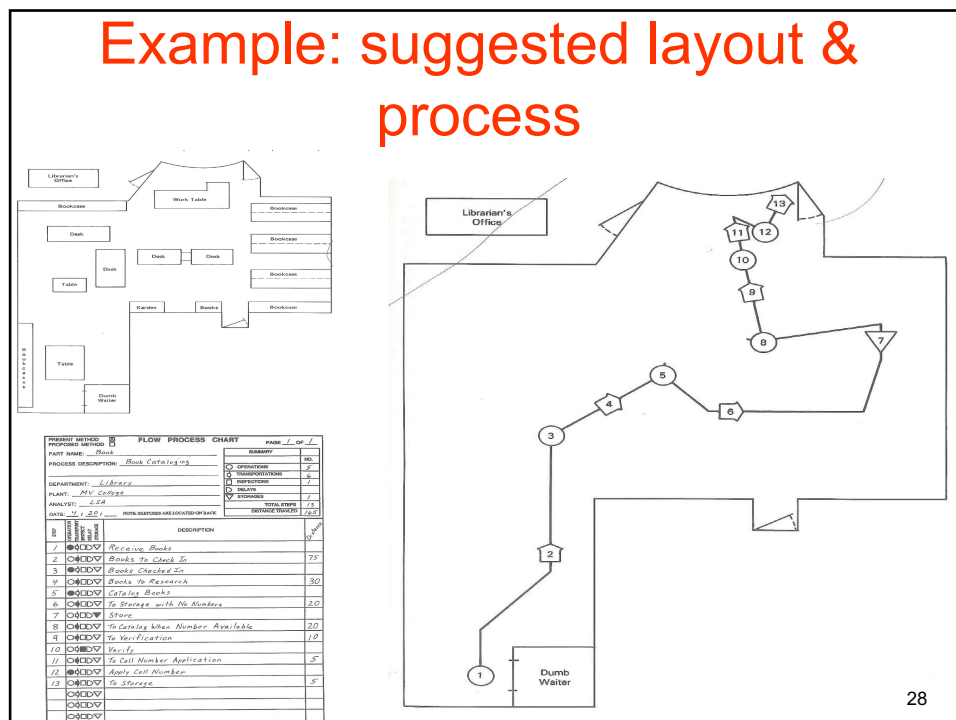
26

Example: compare two layouts



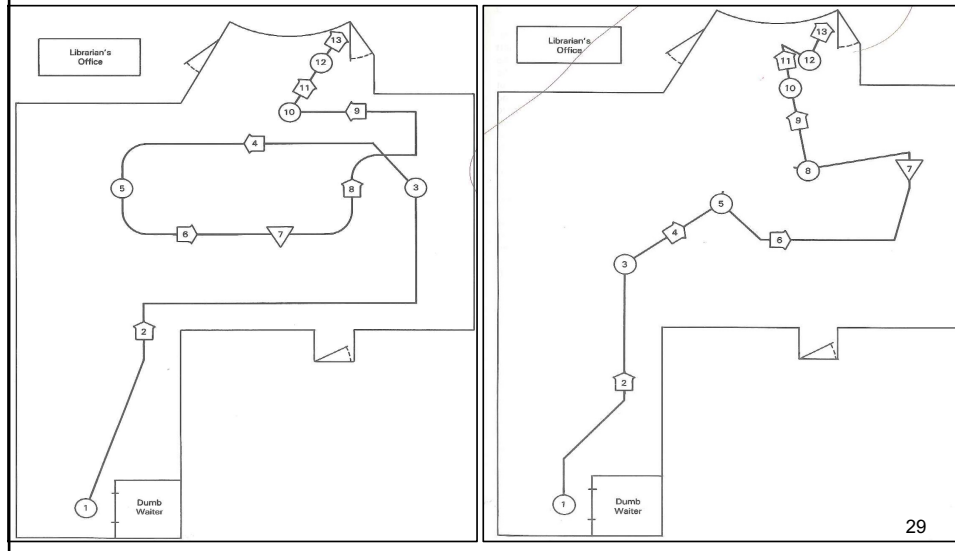
27

Example: suggested layout & process



28

Example: compare two processes



29

Describing processes visually

- Process charts: Operations process chart & Flow process chart (Man, Machine, Material)

– Symbols

- Operation 
- Storage 
- Inspection 
- Delay (flow process chart only) 
- Transportation (flow process chart only) 

INDR 100

30

30

Class exercise

- Draw a flow process chart for the preparation of a strawberry pie. The recipe is given below (courtesy of Mr. David F. Byrd):

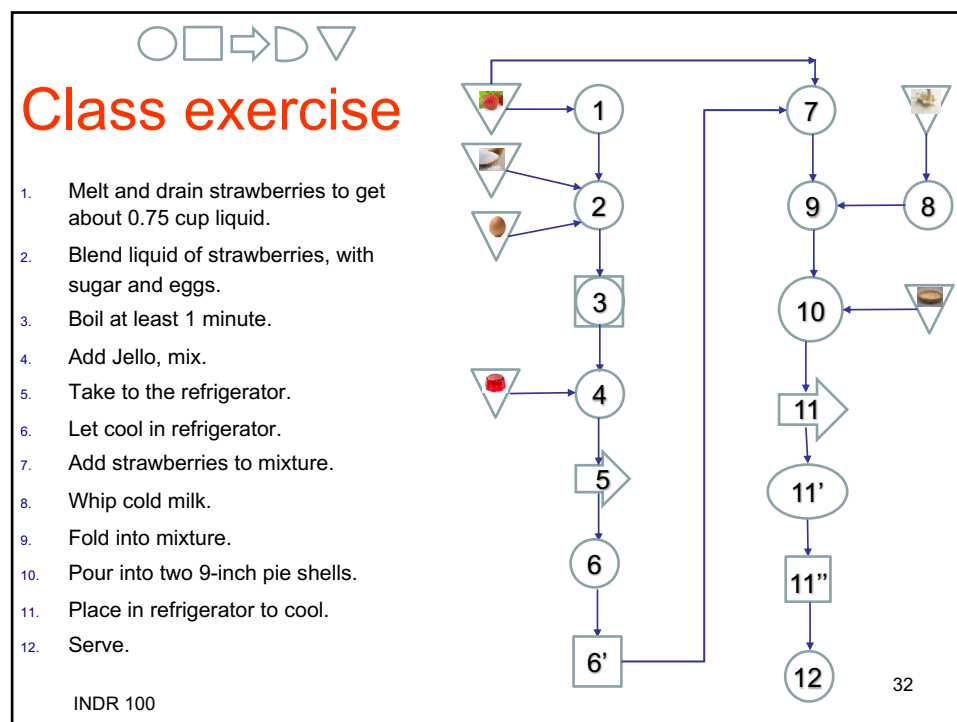
1. Melt and drain strawberries to get about 0.75 cup liquid.
2. Blend liquid of strawberries, with sugar and eggs.
3. Boil at least 1 minute.
4. Add Jello, mix.
5. Take to the refrigerator.
6. Let cool in refrigerator.
7. Add strawberries to mixture.
8. Whip cold milk.
9. Fold into mixture.
10. Pour into two 9-inch pie shells.
11. Place in refrigerator to cool.
12. Serve.

Ingredients

1 cup sugar
1 package frozen strawberries (1.5 cups)
2 well-beaten eggs
1 package strawberry Jello
1 can evaporated milk

31

31



32